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SUMMARY

Master of Computer Science graduate from the University of Sydney (9 course distinctions) and BSc graduate from Bilkent University (Magna Cum Laude), with strong foundations in algorithms, machine learning, cybersecurity, and distributed systems, complemented by excellent problem-solving and analytical skills.

EDUCATION

University of Sydney, Sydney, NSW, Australia — Master of Science in Computer Science
WAM 75%, Feb 2024 - Feb 2026
System Programming – Data Engineering – Information Theory and Complex Systems – Applied Cybersecurity – Natural Language Processing – Cybersecurity Engineering – Parallel and Distributed Computing – Large Scale Networks

Bilkent University, Cankaya, Ankara, Turkiye — Bachelor of Science in Computer Engineering
CGPA: 3.60/4, Graduated: Magna cum Laude, Top 10 percentile. Duration: Sep 2019 – Jul 2023.
Algorithms II (CS474 or CS502) – Statistical Learning & Data Analysis – Operating Systems – Probability & Statistics – Database Systems – Neural Networks – Introduction to Machine Learning – Estimation & Detection Theory

WORK HISTORY

Software Engineer — May 2026 to Present
3 Tier Technology (3TT), Hobart, Tasmania, Australia
- Delivered 8 major features and 14 bug fixes to Casey, a web application used by Anglicare, using TSQL, Visual Basic, and ASPX, including corrections to historic data to restore data integrity.
- Upgraded the PAB (Property Agents Board) web application to support granting and sending licences, improving data integrity across licence records, and introducing new assessment mechanisms .
- Built and maintained the EDI2 application (used by Inghams to receive and process customer orders) using C#, MySQL, and ASP.NET. Delivered 26 endpoints, 2 services, internal logging and auditing mechanisms, and 10+ user interfaces, in addition to file-processing, client-information-maintenance, and security features, and fixed 9+ bugs.
- Wrote database procedures and views and added new tables across the Casey, PAB, and EDI2 applications to support feature delivery and data integrity.

Backend Developer (Graduate Project) — Aug 2025 to Dec 2025
RenoPilot Pty Ltd (Remote), Australia
- Worked with a team to ship features for RenoPilot, an Australian proptech startup.
- Built 18+ RESTful API endpoints and connected the backend and frontend for a cloud-based roof replacement app.
- Ran unit and integration tests to ensure end-to-end reliability and code quality.
- Validated backend data interactions across multiple schemas in a PostgreSQL database.

Technology Consultant — Jul 2023 to Mar 2024
EnCon Asia Consultants, J&K, India
- Architected and launched the company's web platform from the ground up, managing software deployment across devices for a team of 10.
- Conducted a digital infrastructure audit and managed digital infrastructure.
- Integrated technology into workflows to utilize technology in sustainable smart city design.

Software Engineering Intern — Jun 2022 to Aug 2022
Bayzat (Ankara Office), Dubai, UAE
- Built a GoLang-based software management tool to automate infrastructure and deployment changes across an organisation of 200-500 employees.
- Took part in application architecture design and implemented a Text User Interface (TUI) using the Bubble Tea and Glow libraries.
- Developed frontend logic and interface components for the application and mentored a junior intern.

Software Engineering Intern — Sep 2021 to Dec 2021
Radity (Remote), Zurich, Switzerland
- Built and maintained 3 Python/Django packages for a SaaS application, improving delivery speed across the team.
- Set up SMTP email integration using Django to support core application functionality.
- Spotted 10+ security vulnerabilities on a production website through a targeted Metasploit investigation and other penetration testing tools.

Software Engineering Intern — Jun 2021 to Sept 2021
FDNSOFT, Ankara, Turkiye
- Added an advanced search feature and improved graphics for a large-scale production application.
- Added 3 major UI updates, including animation, to the previous version and contributed to UI design for new interfaces.
- Identified and fixed 13 bugs in the Flutter frontend and 3 in the Node.js backend.

CERTIFICATIONS

- CC certified

HONORS

- 9 Course Distinctions @ USYD, Sydney International Student Award @ USYD
- Graduated magna cum laude from Bilkent University, 5 High Honors Lists and 3 Honors Lists @ BILKENT UNIVERSITY
- Bilkent Undergraduate Scholarship 2019-23
- 2nd Rank - 2014 Inter School Essay Competition, Tyndale Biscoe & Mallinson Educational Society

VOLUNTEER & ORGANISATIONS

- **Customer Service** @ Red Cross Society Store, (Mar 2025 to now) Hobart, Tasmania, Australia
  - Assisted 20+ customers regularly with product selection and enquiries, maintaining a high standard of service.
  - Operated the POS system to process sales and handle transactions accurately.
  - Worked alongside a diverse team of 5+ volunteers and staff to keep daily store operations running smoothly.
- **Organising Committee Member**, Australian Kashmiri Association Inc (2024 to now), Sydney, NSW, Australia: Planned and managed events for a community of around 200 members.
- **Organising Committee Member**, Effective Altruism University of Sydney (Aug 2024 to Dec 2024), NSW, Australia: Handled funding applications and organised chapter events.

LANGUAGES

English (excellent verbal and written communication skills: 90/90 in proficiency test), Kashmiri, Urdu, Turkish, basic Spanish

HOBBIES

Running, Reading, Football, Table Tennis, Classic Literature, Poetry (both Urdu and English)

SKILLS

Programming Languages
- C, C++, Python, C#, Java, JavaScript, Go, SQL, Bash

Security & Systems Tools
- ZAPROXY, Burp Suite, Nmap, Valgrind, Ghidra, Apktool, Docker, Azure Sentinel, OpenClaw

Cloud,Platforms, & Operating Systems
- Linux (Ubuntu, Kali), macOS, Windows , Azure, Relational Databases, NoSQL, M

Libraries & Technical Frameworks
- PyTorch, TensorFlow, NumPy, Pandas, NetworkX, Keras, Pthreads, MPI (C++), JDBC, Spring Boot, Express.js, Flask

PROJECTS

Vulnerability Assessment (Applied Cybersecurity)

- Gained root access to a vulnerable VM by exploiting an insecure boot configuration and reset credentials using passwd.
- Exploited multiple attack vectors including FTP anonymous access, Werkzeug debug console exposure, and SQL injection to obtain database credentials and remote shell access.
- Spotted and confirmed web vulnerabilities including XSS, CSRF, and clickjacking, showcasing exploitability through session hijacking and CSRF escalation.
- Uncovered 16 vulnerabilities across network, OS, web server, and database layers using nmap, sqlmap, ZAPROXY, Nikto, Metasploit, and Wireshark; documented findings and remediation in a structured security report.
- Received 100% score in the project.

Democratic National Convention Hack - Incident Review

- Investigated the 2016 DNC cyberattack, identifying the attack vectors, tools, and techniques used by the threat actors.
- Studied the security failures that enabled the breach, including spear-phishing and credential theft.
- Authored a structured incident review report covering the timeline, impact, and recommended mitigations.
- Presented findings to the class, drawing on real-world cybersecurity threats and defence strategies.

Catch The Flags (Applied Cybersecurity)

- Completed 7 out of 8 CTF challenges across reverse engineering, cryptography, and network security.
- Reverse engineered binary executables using Ghidra, Frida, and apktool to analyse program logic and extract flags.
- Wrote custom Python scripts for automated password cracking and cryptographic attacks.
- Carried out a Man-in-the-Middle (MITM) attack to intercept and analyse live network traffic.

LDPC Code Simulation for IEEE 802.16e (WiMAX) - Error Control Coding (ELEC5507)

- Built a full IEEE 802.16e-compliant LDPC coding system in MATLAB, covering encoding, BPSK modulation, SPA-based decoding, and BER/FER evaluation across AWGN and Rayleigh fading channels.
- Applied the Sum-Product Algorithm (SPA) using LLR-based belief propagation on a Tanner graph with 2304 variable nodes and 1152 check nodes.
- Simulated and studied BER/FER performance, showcasing near-zero BER with a ~1.5 dB Eb/N0 increase in AWGN and classic waterfall behaviour under Rayleigh fading up to 60 dB.

CodeBank - Full-Stack Development Project

- Built a discussion and coding problem platform enabling user engagement and coding interview simulations.
- Planned the database and application architecture alongside teammates.
- Shipped features including signup, login, meeting organisation, a discussion forum, and problem-solving challenges.
- Used Spring Boot for the backend, React for the frontend, and SQL for the database.
- Received an A grade; successfully delivered for the Database Systems course at Bilkent University.

Building and Orchestrating a Data Pipeline for an Analytics Suite - Data Engineering

- Built an end-to-end data pipeline using Apache Airflow and dbt, containerised with Docker.
- Set up Airflow DAGs to orchestrate ETL workflows for synthetic and real-world healthcare data (MIMIC).
- Applied staging, dimension, and fact models in dbt with automated tests to ensure data integrity.
- Created interactive dashboards in Apache Superset to analyse and visualise warehouse data.

Lung Disease Classifier - Machine Learning

- Built diverse machine learning models (SVMs, Neural Networks, ViTs) to classify lung X-rays into five categories.
- Compared the performance of different structures on the classification task, including 1v1 and 1vAll SVM approaches.
- Achieved 92% accuracy on the test dataset.
- Earned a 100/100 score for the final report.

Homophily and Giant Connected Component in Stock Time Series Data (Large Scale Networks)

- Processed time-series data for 4,440 stocks over 5 months using multiple financial APIs.
- Built stock similarity graphs using entropy-based metrics including Pointwise Mutual Information (PMI) and Mean Absolute Distance (MAD).
- Studied graph properties such as sparsity, clustering coefficient, connectivity, and centrality to explore sector clustering and index fund influence.
- Observed the emergence of a giant connected component, interpreting it as evidence of market sentiment and systemic interdependence.

Breast Cancer Feature Selection — Information-Theoretic Measures

- Compared different mechanisms to select a subset of features that maximised the mutual information about the presence or absence of cancer.
- Used measures such as Mutual Information and Conditional Mutual Information to find the subset of features containing maximum information about the label.
- Trained a PyTorch-based Autoencoder for dimensionality reduction alongside PCA.
- Received a distinction for the project final report.

Election Strategy Optimisation - Integer Linear Programming (IE 400)

- Modelled a multi-part election strategy problem as a series of ILPs, optimising campaign investment across 35 cities on a 5x7 grid under neighbourhood and voting constraints.
- Formulated and solved a minimum-cost investment plan to ensure majority city wins.
- Planned a region partitioning ILP to maximise council seats across 7 connected regions of 5 cities each, subject to border laws and adjacency constraints.
- Extended the partitioning model with graph connectivity constraints, enforcing spatial contiguity via domino-based neighbour forcing across a 5x5 inner grid.
- Ran all formulations in Gurobi and confirmed optimal solutions computationally.